

General Relativity Exercise 6

Idan Stark

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1 Warmup

Starting from the Einstein equation

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = T_{\mu\nu} \quad (1)$$

Taking the trace

$$R - \frac{1}{2}Rg = T \quad (2)$$

Solving for R :

$$R = \frac{2}{1-g}T \quad (3)$$

Assigning this back into the original Einstein's equation and solving for $R_{\mu\nu}$ we get

$$R_{\mu\nu} = T_{\mu\nu} + \frac{g_{\mu\nu}}{1-g}T \quad (4)$$

2 Newtonian Approximation in Lensing

Let us consider the point mass lens with weak Newtonian gravitational potential with the following metric:

$$ds^2 = \left(1 + \frac{2\Phi}{c^2}\right) c^2 dt^2 - \left(1 - \frac{2\Phi}{c^2}\right) d\vec{x}^2 \quad (5)$$

Where Φ is the Newtonian gravitational potential generated by a point mass M :

$$\Phi(\vec{x}) = -\frac{GM}{r} \quad (6)$$

2.1 The deflection angle

According to Fermat's principle, light travels along geodesics. So, let us find the geodesics of the above metric. In spherical coordinates, the metric and inverse metric are:

$$\begin{aligned} g_{\mu\nu} &= \text{diag} \left(1 + \frac{2\Phi}{c^2}, -\left(1 - \frac{2\Phi}{c^2}\right), -\left(1 - \frac{2\Phi}{c^2}\right)r^2, -\left(1 - \frac{2\Phi}{c^2}\right)r^2 \sin^2 \theta \right) \\ g^{\mu\nu} &= \text{diag} \left(\frac{1}{1 + \frac{2\Phi}{c^2}}, \frac{-1}{1 - \frac{2\Phi}{c^2}}, \frac{-1}{\left(1 - \frac{2\Phi}{c^2}\right)r^2}, \frac{-1}{\left(1 - \frac{2\Phi}{c^2}\right)r^2 \sin^2 \theta} \right) \end{aligned} \quad (7)$$

This gives the Christoffel symbols:

$$\begin{aligned}
\Gamma_{tt}^r &= \frac{\partial_r \Phi}{c^2 - 2\Phi} & \Gamma_{rr}^r &= -\frac{\partial_r \Phi}{c^2 - 2\Phi} & \Gamma_{tr}^t &= \Gamma_{rt}^t = \frac{\partial_r \Phi}{c^2 + 2\Phi} \\
\Gamma_{\theta\theta}^r &= r - \frac{\partial_r \Phi}{c^2 - 2\Phi} r^2 & \Gamma_{\varphi\varphi}^r &= \Gamma_{\theta\theta}^r \sin^2 \theta \\
\Gamma_{\theta r}^\theta &= \Gamma_{r\theta}^\theta = \Gamma_{\varphi r}^\varphi = \Gamma_{r\varphi}^\varphi = \frac{1}{r} - \frac{\partial_r \Phi}{c^2 - 2\Phi} \\
\Gamma_{\varphi\varphi}^\theta &= -\frac{1}{2} \sin 2\theta & \Gamma_{\varphi\theta}^\varphi &= \Gamma_{\theta\varphi}^\varphi = \cot \theta
\end{aligned} \tag{8}$$

Then, the geodesic equation

$$\ddot{x}^\mu + \Gamma_{\alpha\beta}^\mu \dot{x}^\alpha \dot{x}^\beta = 0 \tag{9}$$

where $\dot{x}$ is the derivative by a parameter λ , becomes:

$$\begin{cases}
c\ddot{t} + \frac{2\partial_r \Phi}{c^2 + 2\Phi} \dot{r} c \dot{t} = 0 \\
\ddot{r} + \frac{\partial_r \Phi}{c^2 - 2\Phi} c^2 \dot{t}^2 - \frac{\partial_r \Phi}{c^2 - 2\Phi} \dot{r}^2 + \left(r - \frac{\partial_r \Phi}{c^2 - 2\Phi} r^2 \right) (\dot{\theta}^2 + \sin^2 \theta \dot{\varphi}^2) = 0 \\
\ddot{\theta} - \frac{1}{2} \sin 2\theta \dot{\varphi}^2 + 2 \left(\frac{1}{r} - \frac{\partial_r \Phi}{c^2 - 2\Phi} \right) \dot{r} \dot{\theta} = 0 \\
\ddot{\varphi} + 2 \left(\frac{1}{r} - \frac{\partial_r \Phi}{c^2 - 2\Phi} \right) \dot{r} \dot{\varphi} + 2 \cot \theta \dot{\theta} \dot{\varphi} = 0
\end{cases} \tag{10}$$

Since the motion is in a plain, we can set $\theta = \pi/2$ and so $\dot{\theta} = 0$. The third equation vanishes and we are left with:

$$\begin{cases}
c\ddot{t} + \frac{2\partial_r \Phi}{c^2 + 2\Phi} \dot{r} c \dot{t} = 0 \\
\ddot{r} + \frac{\partial_r \Phi}{c^2 - 2\Phi} c^2 \dot{t}^2 - \frac{\partial_r \Phi}{c^2 - 2\Phi} \dot{r}^2 + \left(r - \frac{\partial_r \Phi}{c^2 - 2\Phi} r^2 \right) \dot{\varphi}^2 = 0 \\
\ddot{\varphi} + 2 \left(\frac{1}{r} - \frac{\partial_r \Phi}{c^2 - 2\Phi} \right) \dot{r} \dot{\varphi} = 0
\end{cases} \tag{11}$$

From the first equation, defining $v_t = c\dot{t}$:

$$\begin{aligned}
\dot{v}_t + \frac{2\partial_r \Phi}{c^2 + 2\Phi} \dot{r} v_t &= 0 \\
\frac{dv_t}{v_t} &= -\frac{2\partial_r \Phi}{c^2 + 2\Phi} dr \\
\frac{dv_t}{v_t} &= -\frac{2}{c^2 + 2\Phi} d\Phi \\
v_t &= \alpha \frac{1}{c^2 + 2\Phi}
\end{aligned} \tag{12}$$

This gives:

$$\begin{cases}
\ddot{r} + \frac{\partial_r \Phi}{c^2 - 2\Phi} \alpha^2 \frac{1}{(c^2 + 2\Phi)^2} - \frac{\partial_r \Phi}{c^2 - 2\Phi} \dot{r}^2 + \left(r - \frac{\partial_r \Phi}{c^2 - 2\Phi} r^2 \right) \dot{\varphi}^2 = 0 \\
\ddot{\varphi} + 2 \left(\frac{1}{r} - \frac{\partial_r \Phi}{c^2 - 2\Phi} \right) \dot{r} \dot{\varphi} = 0
\end{cases} \tag{13}$$

Performing the same manouver for the φ equation, we can write $\omega = \dot{\varphi}$ and get:

$$\begin{aligned}
\dot{\omega} + 2 \left(\frac{1}{r} - \frac{\partial_r \Phi}{c^2 - 2\Phi} \right) \dot{r} \omega &= 0 \\
\frac{d\omega}{\omega} &= -2 \left(\frac{dr}{r} - \frac{d\Phi}{c^2 - 2\Phi} \right) \\
\omega &= \beta \frac{1}{r^2 (c^2 - 2\Phi)}
\end{aligned} \tag{14}$$